

main.c

Run

```
1 #include <stdio.h>
2 int main()
3 {
4     int pages[50], frames[10], recent[10];
5     int n, f, i, j, k;
6     int pageFault = 0;
7     int time = 0;
8     int found, pos, least;
9     printf("Enter number of pages: ");
10    scanf("%d", &n);
11    printf("Enter page reference string:\n");
12    for(i = 0; i < n; i++)
13    {
14        scanf("%d", &pages[i]);
15    }
16    printf("Enter number of frames: ");
17    scanf("%d", &f);
18    for(i = 0; i < f; i++)
19    {
20        frames[i] = -1;
21        recent[i] = 0;
22    }
23    printf("\nPage\tFrames\n");
24    for(i = 0; i < n; i++)
25    {
26        found = 0;
27        time++;
28        for(j = 0; j < f; j++)
29        {
30            if(frames[j] == pages[i])
31            {
32                found = 1;
33                recent[j] = time;
34                break;
35            }
36        }
37        if(found == 0)
38        {
39            pageFault++;
40            pos = -1;
41            for(j = 0; j < f; j++)
42            {
43                if(frames[j] == -1)
44                {
45                    pos = j;
46                    break;
47                }
48            }
49            if(pos == -1)
50            {
51                least = recent[0];
52                pos = 0;
53                for(j = 1; j < f; j++)
54                {
55                    if(recent[j] < least)
```

Output

Enter number of pages: 10
Enter page reference string:
1 2 3 1 4 5 2 1 2 3
Enter number of frames: 3

Page	Frames
1	1 - - Fault
2	1 2 - Fault
3	1 2 3 Fault
1	1 2 3 Hit
4	1 4 3 Fault
5	1 4 5 Fault
2	2 4 5 Fault
1	2 1 5 Fault
2	2 1 5 Hit
3	2 1 3 Fault

Total Page Faults = 8

=== Code Execution Successful ===